

ISYE 6420 Homework 3

Question 1

1a)

To elicit a beta prior, we can use the existing knowledge of the expected value and variance to determine α and β

- Beta PDF: $f(x|\alpha, \beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$
- Beta Mean: $\frac{\alpha}{\alpha+\beta}$
- Beta Variance: $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

We know $\mu = 0.9$ and $\mu - 2\sigma = 0.8$

$$\sigma = \frac{\mu - 0.8}{2} = \frac{0.9 - 0.8}{2} = \frac{1}{20} = 0.05$$

$$\mu = \frac{\alpha}{\alpha + \beta} = 0.9$$

$$\alpha = \frac{9}{10}(\alpha + \beta)$$

Hence:

$$\alpha = 9\beta$$

$$\sigma^2 = \frac{\alpha}{\alpha + \beta} \cdot \frac{\beta}{(\alpha + \beta)(\alpha + \beta + 1)} = \mu \cdot \frac{\beta}{10\beta(10\beta + 1)} = \frac{9}{10} \frac{\beta}{10\beta(10\beta + 1)} = \left(\frac{1}{20}\right)^2$$

$$\frac{\beta}{\beta(10\beta + 1)} = \left(\frac{1}{400}\right)\left(\frac{100}{9}\right)$$

$$\frac{1}{10\beta + 1} = \frac{1}{36}$$

$$\beta = \frac{36 - 1}{10} = 3.5$$

$$\alpha = 9\beta = 9(3.5) = 31.5$$

Therefore the Beta prior:

$$\pi(\theta) \sim B(31.5, 3.5)$$

1b)

For the conjugate case of a Beta prior, the likelihood will follow a Binomial Distribution:

$$X|\theta \sim \text{Bin}(n = 30, \theta)$$

We can use the kernel method to see the posterior update of parameters for our conjugate family.

$$f(X|\theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k} \text{ and } \pi(\theta) = \frac{\theta^{\alpha-1} (1-\theta)^{\beta-1}}{B(\alpha, \beta)}.$$

$$\pi(\theta|X) \propto f(X|\theta) \cdot \pi(\theta)$$

$$\theta^{\alpha-1} (1 - \theta)^{\beta-1} \propto f(X|\theta) = (\theta^k (1 - \theta)^{n-k}) \cdot (\theta^{\alpha-1} (1 - \theta)^{\beta-1}) = \theta^{(\alpha-1)+k} \theta^{(\beta-1)+(n-k)}$$

Posterior update of parameters given that k, number of success, is 22:

$$\alpha_{post} = \alpha_{prior} + k = 31.5 + 22 = 53.5$$

$$\beta_{post} = \beta_{prior} + (n - k) = 3.5 + (30 - 22) = 11.5$$

$$\pi(\theta|X) \sim B(53.5, 11.5)$$

$$E(\theta|X) = \frac{\alpha}{\alpha + \beta} = \frac{53.5}{65} \approx 0.82$$

1c)

The pdf must evaluate to the same density at each bound (so k is horizontal).

The probability within those bounds must be equal to 1 - alpha, where alpha = 0.05 for our 95% HDI credible set.

Solving numerically:

- HDI Credible Interval: [0.73, 0.91]
- Probability within set: 0.95
- Length of interval: 0.18

1d)

$$p_0 = \int_{0.8}^{\inf} \pi(\theta|x) d\theta$$

$$p_1 = \int_{\inf}^{0.8} \pi(\theta|x) d\theta$$

Use the CDF of the Beta function to evaluate the two hypotheses and their associated integrals, producing:

$$p_0 = 0.71 \text{ and } p_1 = 0.29$$

Question 2

Find the posterior distribution using Morris MLE.

Assuming the prior also has a normal distribution:

$$\pi(\theta|\mu) \sim N(\mu, \tau^2)$$

$$f(X|\theta) \sim N(\theta, \sigma^2 = (0.02)^2)$$

First we find the marginal likelihood for x_i

$$m_i(x_i|\mu) = \int f_i(x_i, \theta_i) \pi(\theta|\mu) d\theta_i$$

$$m(x|\mu) = \prod_{i=1}^n m_i(x_i|\mu)$$

Since X_i distributions are all independent and identically distributed, the variances can simply be added. So under empirical bayes, we can say the marginal distribution follows:

$$X_i \sim_{iid} N(\mu, \sigma^2 + \tau^2)$$

Hence the marginal likelihood can be summarised with the pdf:

$$m(x|\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi(\sigma^2 + \tau^2)}} e^{\frac{-(x_i - \mu)^2}{2(\sigma^2 + \tau^2)}}$$

Using the MLII method, we can say the best estimate for μ is the sample mean $\bar{X}$. Similarly the best estimate for variance is the sample variance,

The sample mean can be calculated:

$$\bar{X} = \frac{3.29 + 3.31 + 3.35 + 3.34 + 3.28}{5} = 3.314$$

Sample variance is a combination of our prior and likelihood variances, $s^2 = \sigma^2 + \tau^2$, which we can also calculate directly:

$$s^2 = \frac{\sum (x_i - \bar{X})^2}{n - 1} = 0.0009$$

Posterior mean and variance update for Normal-Normal conjugate families are:

$$\mu_{post} = \frac{\sigma^2 \mu + \tau^2 x}{\tau^2 + \sigma^2} = \hat{B} \mu + (1 - \hat{B}) x$$

$$\tau_{post}^2 = \hat{B} \tau^2 = (1 - \hat{B}) \sigma^2$$

where

$$\hat{B} = \frac{\sigma^2}{\tau^2 + \sigma^2} = \frac{\sigma^2}{s^2} = \frac{0.02^2}{0.0009} = 0.43$$

In our case, to find θ_6 distribution given $x_6 = 3.30$:

$$\hat{\theta}_6 = \hat{B}\bar{X} + (1 - \hat{B})x_6 = 0.43(3.314) + (1 - 0.43)(3.3) = 3.306$$

The variance of θ_6 can be calculated:

$$\tau_6^2 = (1 - \hat{B})\sigma^2 = 0.000228$$

Numerically determining the 95% HDI Credible Set based on the updated posterior distribution $\sim N(\hat{\theta}_6, \tau_6^2)$:

- HDI Credible Interval: [3.28, 3.34]
 - Probability within set: 0.95
 - Width of interval: 0.06
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Question 3

Parasol mushroom distribution $\sim N(\mu, 9)$

Likelihood: $f(x|\mu = 17, n = 10) \sim N(17, \frac{\sigma}{10})$

Prior: $\pi(x) \sim N(18, \sigma^2)$

As we are testing for a precise null in a continuous distribution, we need to use a point mass.

$$\pi(\mu) = \pi_0\delta_{18} + \pi_1\xi(\mu) = 0.8\delta_{18} + 0.2\xi(\mu)$$

where $\pi_0 + \pi_1 = 1$.

To determine the posterior distribution first the marginal has to be found.

$$\pi(\mu_0|y) = \frac{f(y|\mu_0)\pi_0}{m(y)}$$

$$m(y) = \pi_0 \cdot f(y|\mu_0) + \pi_1 m_1(y)$$

Where:

$$m_1(y) = \int_{\mu \neq \mu_0} f(y|\mu)\xi(\mu) \cdot d\mu$$

$$\begin{aligned}\pi(\mu_0|y) &= \frac{\pi_0 f(y|\mu_0)}{\pi_0 f(y|\mu_0) + \pi_1 m_1(y)} \\ &= \frac{\pi_0 f(y|\mu_0)}{\pi_0 f(y|\mu_0) \left(1 + \frac{\pi_1 m_1(y)}{\pi_0 f(y|\mu_0)}\right)} \\ &= \frac{1}{\left(1 + \frac{\pi_1 m_1(y)}{\pi_0 f(y|\mu_0)}\right)}\end{aligned}$$

$$\pi(\mu_0|y) = \left(1 + \frac{\pi_1}{\pi_0} \cdot \frac{m_1(y)}{f(y|\mu_0)}\right)^{-1}$$

$$B_{01} = \frac{f(y|\mu_0)}{m_1(y)}, \quad B_{10} = \frac{m_1(y)}{f(y|\mu_0)}$$

For evaluating $m_1(x)$, we can use the property of normal distributions (as both the likelihood and spread are normally distributed) for two independent normally distributions.

$$m_1(y) \sim N(\mu, \sigma_L^2 + \tau_\xi^2)$$

where $f(y|\mu) \sim N(\mu, \frac{\sigma^2}{n})$ and $\xi(\mu) \sim N(\mu, \tau_\xi^2)$.

We know the likelihood has variance $\frac{\sigma^2}{N} = \frac{9}{10}$ and from the given formula for the spread we can infer that it has variance, $\tau_\xi^2 = 12$. Therefore:

$$m_1(y) \sim N(\mu, 12.9)$$

$$B_{10} = \frac{m_1(y)}{f(y|\mu_0)} \Big|_{y=17}^{\mu=18} = \frac{m_1(17)}{f(17, 18)} = 0.443$$

Evaluating the Bayes Factor with the Jeffrey's Scale:

$$\log_{10} B_{01} = -\log_{10} B_{10} = -\log_{10}(0.443) = 0.354$$

The Bayes Factor appears as Grade 1 on the Jeffrey's scale which supports the null hypothesis and evidence for H_1 is barely worth mentioning.

Calculating posterior probability:

$$p_0 = P(H_0|y) = \frac{f(y|\mu_0)}{m(y)} = \frac{f(y|18)}{0.8f(y|18) + 0.2m_1(y)} = \left(1 + \frac{0.2}{0.8} \frac{m_1(y)}{f(y|18)}\right)^{-1} = (1 + 0.25(B_{10}))^{-1} = 0.9$$

$$p_1 = P(H_1|y) = 1 - p_0 = 1 - 0.9 = 0.1$$